Fundamentals of Machine Learning

Tutorial 1, Extra Practice: Problems

From learning problems to probabilistic classification. Same skills as Tutorial 1, new settings and numbers.

How to use this set

Eight problems, one for each problem of Tutorial 1, in the same order and with the same skills: formulating a learning problem, decisions from costs, the geometry of KNN, auditing an evaluation, capacity and selection, discriminative probabilities, generative models, and detection over a sequence. The settings and the numbers are new, and each problem adds one twist the tutorial did not ask. Part B is ten two-minute drills for speed.

Each problem opens with a blue box holding the setup and a suggested time, then a yellow box listing the tools you need with a pointer [one-read §x] to the one-read document. Work on paper first; the same set with full solutions is *ML_Tutorial1_Extra_Practice_Solutions.pdf*. Difficulty: ⋆ direct application, ⋆⋆ several steps, ⋆⋆⋆ derivation or proof. All logarithms are natural.

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1 Part A: Tutorial-style problems

1.1 Problem 1: From a phishing objective to a learning problem

Problem 1 (⋆) **15 minutes**

A company's mail gateway receives every incoming email before delivery. The raw record contains the headers (sender and reply-to addresses, routing hops, SPF/DKIM/DMARC authentication results), the subject, the body, the links and the attachments. Labels come from triage: when users report an email, an analyst marks it *phishing* or *legitimate*. When an analyst confirms a campaign, every email sent from the same sending infrastructure is also labelled phishing, even if that particular email was never reported. The deployed system must score each incoming email and decide whether to *deliver* it, *deliver it with a warning banner*, or *quarantine* it.

Tools you need

The pipeline object → representation → target → model output → action [one-read §1.4, 1.5]; the posterior versus the decision [one-read §2.2]; leakage and shortcuts [one-read §3.6]; the three learning settings [one-read §1.2].

(a) Identify the physical object, the representation x, the target Y, the model output, and the final action.

(b) A legitimate password-reset email from the company's IT provider and a phishing email cloned from it can have almost identical observed features. Explain why a classifier using only these features may be unable to separate them. Does this mean the labels are wrong?

(c) Explain why a posterior estimate $\hat{p}_\theta(y = 1 \mid x)$ and the quarantine decision are not the same object.

(d) For each candidate feature, say whether it is available at decision time and whether it would create target leakage: (i) the number of recipients who reported the email; (ii) whether the sender domain was registered less than 30 days before arrival; (iii) whether a link in the email was later added to a blocklist; (iv) the DKIM result.

(e) In the collected data, the phishing emails were exported from a quarantine archive that converts every message to plain text, while the legitimate emails were exported from user mailboxes in their original HTML. Explain how "the email contains HTML formatting" can become a learning shortcut. Why is this different from target leakage? How could you detect the problem?

(f) Describe one email-security task for each paradigm: supervised, self-supervised, and reinforcement learning. For each, identify the information that provides the learning signal.

1.2 Problem 2: The optimal action depends on the loss

Problem 2 (⋆⋆) **15 minutes**

An endpoint-protection agent must decide what to do with a newly downloaded file. For a file with features x let $p = p(y = 1 \mid x)$, where class 1 means *malicious*. Allowing a malicious file costs $C_{FN} = 12$. Blocking a benign file costs $C_{FP} = 3$, because it interrupts the user's work. Correct decisions cost 0.

Tools you need

Conditional risk of an action: the average loss given x, $R(a \mid x) = \sum_y c(a, y)\, p(y \mid x)$. Bayes rule: take the action with the smallest conditional risk [one-read §2.2].

(a) Derive the conditional risks of *allow* and *block*, and the optimal probability threshold.

(b) Apply the rule to four files with posteriors 0.05, 0.18, 0.25, 0.60.

(c) A sandbox can now detonate the file and observe its behaviour. Sandboxing costs 1 whatever the true class, and then the file is handled correctly. Derive the probability regions for allow, sandbox, and block, and redo (b).

(d) The sandbox is leaky. It costs only 0.5, and benign files are always cleared correctly, but 10% of malicious files detect the sandbox, behave innocently, and are then allowed. Derive the conditional risk of sandboxing, the new regions, and the decisions for the four files.

(e) Back to the perfect sandbox of (c), but with unknown cost c. What is the largest c for which sandboxing is optimal for at least one file? Interpret.

(f) On the build servers, blocking a benign file stops a release and costs 9 instead of 3. Compute the new threshold, and decide the file with $p = 0.25$ in both settings.

1.3 Problem 3: KNN inherits the geometry of the representation

Problem 3 (⋆⋆) 15 minutes

A host is described by $x = (x_1, x_2)$, where x_1 is the number of distinct internal hosts it contacted in the last 10 minutes and x_2 the volume it uploaded to external addresses, in MB. A query host has

$$x = (5,\ 40).$$

Two labelled hosts are $x_A = (5,\ 200)$, labelled *attack* (data exfiltration), and $x_B = (20,\ 50)$, labelled *benign* (a backup server).

Tools you need

Euclidean distance, rescaling $z_j = x_j/c_j$; for 1-NN with two training points, the boundary is the set of points equidistant from both, the perpendicular bisector [one-read §2.1, 2.4].

(a) Using raw Euclidean distance, determine the nearest host.

(b) Scale the host count by 5 hosts and the upload by 100 MB. Determine the nearest host after scaling.

(c) In the scaled (z_1, z_2) plane, derive and plot the 1-NN decision boundary between z_A and z_B. Mark the query and give its predicted class.

(d) Use the weighted squared distance $d_w^2(z, z') = w(z_1 - z_1')^2 + (z_2 - z_2')^2$ with $w > 0$. Derive the critical w at which the nearest host changes, say which host is nearest on each side, and interpret.

(e) A colleague adds a third feature, the destination port as a raw integer (22, 443, 3389, 8080). Why is Euclidean distance on this feature meaningless? Propose a better representation.

(f) The scales 5 and 100 were obtained as the standard deviations of the two features over the whole dataset, test set included. Is this leakage? What if they had been chosen from domain knowledge?

1.4 Problem 4: Audit a generalisation claim

Problem 4 (⋆⋆) **15 minutes**

A team collects 60,000 Windows executables gathered between 2019 and 2025: malware from many families, many of them repacked variants of one another, and benign software including many versions of the same programs. The team proceeds as follows:

- fill missing values with the median computed on the full dataset;
- keep the 200 features most correlated with the label, computed on the full dataset;
- randomly split individual files 80/20 into training and test sets;
- tune hyperparameters by 5-fold cross-validation on the training set, with random folds;
- inspect the misclassified test files, add features that fix them, retrain; repeat 15 times;
- report the final test accuracy as the expected performance on next year's malware.

Tools you need

Three subsets and their roles [one-read §2.7]; selection and leakage [one-read §3.5, 3.6]; precision, recall, F_1, and the base rate [one-read §3.6].

(a) Identify every mechanism that can make the reported performance too optimistic.

(b) Propose a training, validation and test protocol for each goal: (i) new files from the same distribution; (ii) next year's malware; (iii) malware families never seen before.

(c) A test set of 20,000 files contains 400 malicious ones. The classifier gives $TP = 300$, $FP = 600$, $FN = 100$, $TN = 19{,}000$. Compute accuracy, precision, recall, F_1, and the false-positive rate.

(d) Compare with the rule that declares every file benign. Why is accuracy insufficient here?

(e) At deployment only 1 file in 200 is malicious. Assuming the same recall and false-positive rate, compute the precision. What fraction of alerts are false?

(f) State where imputation, feature selection and scaling must be fitted in a cross-validation pipeline.

1.5 Problem 5: Capacity, regularisation and validation

Problem 5 (⋆⋆⋆) **20 minutes**

A logistic regression is trained by minimising $J_\lambda(\theta) = L(\theta) + \lambda\|\theta\|^2$, where L is the average training cross-entropy. Let $\hat{\theta}_\lambda$ be a minimiser. For five values of λ the team records the training loss $L(\hat{\theta}_\lambda)$ (without the penalty) and the validation error rate:

λ	10	1	0.1	0.01	0
training loss $L(\hat{\theta}_\lambda)$	0.52	0.38	0.29	0.24	0.21
validation error	0.27	0.19	0.15	0.17	0.23

Tools you need

Optimality: $J_\lambda(\hat{\theta}_\lambda) \le J_\lambda(\theta)$ for every θ. Capacity, underfitting and overfitting [one-read §3.2]; regularisation [one-read §3.4]; selection noise [one-read §3.5]. For $X \sim \mathcal{N}(\mu, s^2)$: $\mathbb{E}|X| =$

$s\sqrt{2/\pi}\,e^{-\mu^2/(2s^2)} + \mu\,(1 - 2\Phi(-\mu/s))$. Values: $\Phi(0.7071) = 0.7602$, $e^{-0.25} = 0.7788$, $\sqrt{2/\pi} = 0.7979$.

(a) Let $\lambda_1 > \lambda_2 \geq 0$, and write $\theta_1 = \hat{\theta}_{\lambda_1}$, $\theta_2 = \hat{\theta}_{\lambda_2}$. Prove that $\|\theta_1\|^2 \leq \|\theta_2\|^2$ and $L(\theta_2) \leq L(\theta_1)$: weaker regularisation can never increase the training loss.

(b) Does (a) guarantee that the training *error rate* also decreases as λ decreases? Explain.

(c) Which λ would you select? Diagnose the others.

(d) The data grow from 2,000 to 200,000 training examples. Should the team keep $\lambda = 0.1$? Explain.

(e) Two configurations have population errors $R_1 = 0.20$ and $R_2 = 0.22$. Their validation errors are $\widehat{R}_k = R_k + \varepsilon_k$ with $\varepsilon_1, \varepsilon_2$ independent $\mathcal{N}(0, \sigma^2)$ and $\sigma = 0.02$. The configuration with the smaller validation error is selected. Compute (i) the probability of selecting the worse configuration, (ii) the expected reported validation error, and (iii) the expected population error of the selected configuration. Interpret.

(f) The team now tries 100 configurations, all with population error 0.20 and the same σ. Given that the expected maximum of 100 independent standard normals is about 2.51, estimate the expected reported validation error of the winner.

1.6 Problem 6: Discriminative probabilities, decisions and prior shift

Problem 6 (⋆⋆) **20 minutes**

A login service scores each login attempt for account takeover. A logistic model produces a score $s_\theta(x)$ and the posterior estimate

$$r_t(x) = \hat{p}_\theta(y = 1 \mid x) = \sigma(s_\theta(x)) = \frac{1}{1 + e^{-s_\theta(x)}},$$

where class 1 means *takeover*. The available actions are *allow* and *challenge* (ask for a second factor).

Tools you need

Sigmoid and logit, $\operatorname{logit}(p) = \ln \frac{p}{1-p}$; cross-entropy $\ell = -[y \ln r + (1 - y) \ln(1 - r)]$ [one-read §4.2]; cost threshold [one-read §2.2]; prior-shift correction [one-read §4.7]:

$$r_d(x) = \frac{(\pi_d/\pi_t)\, r_t(x)}{(\pi_d/\pi_t)\, r_t(x) + \frac{1-\pi_d}{1-\pi_t}\,[1 - r_t(x)]}.$$

(a) For $s_\theta(x) = -0.4$, compute the posterior estimate, and the cross-entropy loss if the true label is 0 and if it is 1.

(b) Show that minimising the empirical cross-entropy is equivalent to maximising the conditional likelihood of the labels under an i.i.d. assumption. Then show that $\partial \ell / \partial s = r - y$, and interpret the sign.

(c) A needless challenge costs $C_{FP} = 1$ and a missed takeover costs $C_{FN} = 9$. Determine the threshold on the posterior and on the score, and the decision for the login of (a).

(d) For the two-feature model $s_\theta(x) = -4 + x_1 + 2x_2$, plot the decision boundary for threshold 0.5 and for the threshold of (c). Indicate the challenge region and explain the movement.

(e) The model was trained on a balanced sample ($\pi_t = 0.5$) obtained by discarding most legitimate logins, while in deployment $\pi_d = 0.02$. Assuming prior shift and a calibrated $r_t(x) = 0.9$, compute $r_d(x)$.

(f) Show that the correction is a constant shift of the score, $s_d = s_t + \ln \frac{\pi_d}{1-\pi_d} - \ln \frac{\pi_t}{1-\pi_t}$. Compute the shift and check (e).

(g) Combine (c) and (f): the rule "challenge if $r_d \geq 0.1$" is a threshold on the training posterior r_t. Find it. A colleague instead challenges when $r_t \geq 0.5$. Which logins are handled differently, and what does this cost?

(h) Give one advantage and one drawback of having trained on a balanced sample built by discarding most legitimate logins.

1.7 Problem 7: Generative classification, assumptions become visible

Problem 7, Part I: Gaussian class models (⋆⋆) 20 minutes for both parts

Two standardised features $x = (x_1, x_2)^\top$ are modelled as $p(x \mid y = c) = \mathcal{N}(\mu_c, \Sigma)$ with a shared covariance, where

$$\mu_0 = (0,0)^\top, \qquad \mu_1 = (3,0)^\top, \qquad \Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \pi_0 = 0.8, \quad \pi_1 = 0.2.$$

Tools you need

LDA score $s_c(x) = x^\top \Sigma^{-1} \mu_c - \frac{1}{2} \mu_c^\top \Sigma^{-1} \mu_c + \ln \pi_c$; posterior by normalising e^{s_0} and e^{s_1} [one-read §4.4]; the 2×2 inverse $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

(a) Compute Σ^{-1}, then $s_0(x)$ and $s_1(x)$.

(b) Write the boundary $s_0(x) = s_1(x)$ and plot it with the two means. Is it perpendicular to the segment joining the means? Where does it cross that segment?

(c) For $x = (2,0)^\top$, determine the predicted class and the posterior probability of class 1.

(d) Repeat (b) and (c) with equal priors. Explain how the prior moves the boundary while the class models stay the same.

(e) What would change if each class had its own covariance matrix?

Problem 7, Part II: Naive Bayes from counts (⋆⋆)

A training set holds 200 emails: 40 phishing and 160 legitimate. Three binary features: $x_1 = 1$ if some link points to a raw IP address, $x_2 = 1$ if the text contains urgent wording, $x_3 = 1$ if the sender is an internal company address. The number of emails with each feature equal to 1 is:

	x_1 (IP link)	x_2 (urgent)	x_3 (internal sender)
phishing (40)	18	30	0
legitimate (160)	4	32	64

A new email has $x = (1, 1, 1)$: an IP link and urgent wording, sent from an internal account.

Tools you need

Naive Bayes score $\pi_c \prod_j p(x_j \mid c)$, posterior by normalising [one-read §4.5]; Laplace smoothing for a binary feature: $\hat{p}(x_j = 1 \mid c) = (n_{jc} + 1)/(n_c + 2)$.

(a) Using the maximum-likelihood (unsmoothed) estimates, compute the posterior probability of phishing. Why is this dangerous here?

(b) Apply Laplace smoothing. Compute the two unnormalised scores and the posterior. With the cost threshold 0.1 from Problem 6, what is the decision?

(c) Rewrite the computation in odds form, as the prior odds times one likelihood ratio per feature. Which features push towards phishing, and how hard?

(d) Suppose the feature set also contained x_2', "urgent wording in the subject", which in practice almost always equals x_2, with the same estimated probabilities. Recompute the posterior and explain what went wrong.

(e) State the Naive Bayes assumption used, and give one reason why it fails for these email features.

(f) Give one reason to prefer the discriminative route and one reason to prefer the generative route here.

1.8 Problem 8: Detecting a brute-force attack from a sequence of logins

Problem 8 (⋆⋆) 20 minutes

A detector observes SSH login attempts from one source address against a server. Attempt 1 is a legitimate login; a brute-force episode starts at $a_2 = 12$ s. Each row gives the start time a_i, the completion time t_i, the username tried, the outcome, and a per-attempt score s_i from a previously fitted classifier, available at t_i:

i	1	2	3	4	5	6	7
a_i (s)	0	12	20	27	35	41	50
t_i (s)	1	13	21	28	36	42	51
username	alice	root	admin	test	oracle	root	ubuntu
outcome	success	fail	fail	fail	fail	fail	fail
s_i	0.05	0.20	0.25	0.30	0.22	0.35	0.28

The per-attempt detector alerts when $s_i \geq 0.70$. A second detector, at each t_i, forms a window

of length $W = 30$ s ending at t_i: it contains attempt i and every earlier attempt from the same source completed after $t_i - W$. It alerts when the window has *at least 4 attempts*, *at least 4 distinct usernames*, a *failure fraction of at least 0.75*, and a *score sum of at least 0.90*.

Tools you need

Window features computed only from the past; detection delay; grouping keys and leakage-safe splits for sequences [one-read §4.8, 3.6].

(a) Which attempts trigger the per-attempt detector? What does this show?

(b) Compute the four window features after each attempt until the first alert. When is the attack detected? Give the delay from the start of the episode and from the completion of its first attempt.

(c) Repeat with a window of $W = 15$ s. Is an alert raised? Explain the benefit and the limitation of a shorter window.

(d) What is the smallest integer window length W for which the alert is raised at $t = 36$?

(e) State which information is available at each decision time. Give two pieces of information that would introduce future-information leakage.

(f) The source address belongs to the university's VPN gateway, shared by 300 students. How can this affect the window features? Suggest a better grouping key.

(g) Consider discounted accumulation $A_i = 0.5A_{i-1} + s_i$, $A_0 = 0$, with alert threshold 0.55. Compute A_i until it first exceeds the threshold, and compare with (b).

(h) A patient attacker sends one attempt every 20 s, each with score 0.20. Show that the 30-second window detector can never alert. Show that A_i tends to $s/(1-\lambda)$ for $A_i = \lambda A_{i-1} + s$, and find the smallest λ for which the accumulator eventually alerts with threshold 0.55.

(i) Describe a leakage-safe train, validation and test split when several windows are generated from each attack episode.

2 Part B: Quick drills

Two minutes each, no context: the computations that Part A builds on.

(D1) $C_{FP} = 5$, $C_{FN} = 20$. Give the posterior threshold and the corresponding threshold on a logistic score.

(D2) A logistic model outputs $s = 2$ for an example with true label 0. Compute the posterior and the cross-entropy.

(D3) Training prior $\pi_t = 0.3$, deployment prior $\pi_d = 0.1$, calibrated $r_t(x) = 0.5$. Compute $r_d(x)$.

(D4) $TP = 45$, $FP = 15$, $FN = 5$, $TN = 935$. Accuracy, precision, recall, F_1?

(D5) Query $(0, 0)$, training points $A = (3, 3)$ and $B = (0, 4.5)$. Which is nearest under Euclidean distance, and which under Manhattan distance?

(D6) One-dimensional data: 1 (benign), 2 (attack), 2.5 (attack), 4 (benign), 5 (benign). Classify the query 3.2 with 1-NN, 3-NN and 5-NN (majority vote).

(D7) Two configurations with the same population error 0.18, independent validation noise $\sigma = 0.03$. Expected reported error of the winner?

(D8) One feature, $x \mid y = 0 \sim \mathcal{N}(0, 1)$ and $x \mid y = 1 \sim \mathcal{N}(2, 1)$, priors 0.9 and 0.1. Find the Bayes decision threshold on x.

(D9) The word "invoice" appears in 0 of 50 legitimate training emails. Give its MLE and Laplace-smoothed probability for the legitimate class (binary feature).

(D10) An accumulator $A_i = 0.8A_{i-1} + 0.1$ starts at 0, with alert threshold 0.6. Does it ever alert? What discount would be needed?